Dynamics of a Rock-Paper-Scissors Cyclical Competition Model in a 2 Dimensional Cellular Automata

Eduard González Yu–1634355

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1 Abstract

The classic game of Rock-Paper-Scissors serves as a fundamental paradigm for studying evolutionary game dynamics. Beyond its simple rules—where rock beats scissors, scissors beat paper, and paper beats rock—the game defines a cyclical competition model that is instrumental in understanding biological diversity. In nature, these non-hierarchical interactions prevent a single dominant species from exhausting resources, thereby maintaining biodiversity.

In this paper, we investigate the spatial dynamics of RPS and two other variants, within a two-dimensional cellular automata framework. We focus specifically on the self-organized pattern formation, such as entangled spiral waves, that emerges from these local microscopic rules. Furthermore, we analyze how these patterns and species coexistence are influenced by finite-size effects, demonstrating that parameters such as population size and spatial extent are critical in determining the stability of the ecosystem.

2 Introduction

When dealt with conflict, we've all played the classical game of Rock-Paper-Scissors (RPS). It's a simple zero-sum game usually meant to choose between two options.

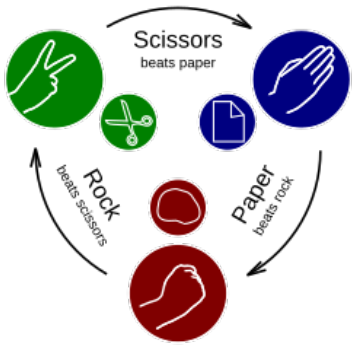

Figure 1: Diagram of how a Rock-Paper-Scissors game is played

It is fair in the sense that there is no winning strategy if the player considers the other players choice to be random.

In nature, this behavior can actually be seen. The *common side-blotched lizard* exhibits a RPS pattern as a mating strategy. The males are divided into three types depending on their throat color; blue, orange and yellow. The different types exhibit different behaviors that compete in an analog way as a RPS game. Orange-throated males are ultradominant and have an advantage over blue-throated individuals but are vulnerable to being a cuckold to a yellow-throated male. Blue-throated lizards are dominant and more able to protect their female from yellow-throated males, but as mentioned, they are dominated by orange-throated ones. Therefore, orange beats blue, blue beats yellow, and yellow beats orange [1].

This form of competition is called cyclic competition. In these models, since no species is dominant the

possibility of coexistence arises. However, this coexistence is dependent on the balance of the species, namely, removing one can lead to instability and loss of biodiversity [2].

We are interested in playing RPS in a grid which we will model using a 2 dimensional cellular automata. A cellular automata (CA) is a model of computation, which consists of a:

- Discrete state space, in this case a 2 dimensional grid which we will represent as a matrix
- Discrete state variables, the collection of possible states (rock, paper or scissors)
- State transition rules, which determine how the grid evolves
- Boundary conditions, how the boundaries behave in accordance with the transition rules
- Initial configuration, how the grid is initialized

The cellular automata will be initialized with a random pattern of rock, paper or scissors states. According to rules we will define in the next section, the state of each cell may change. When running the simulation, we will see how the states interact and the patterns that form from these interactions.

We will also analyze two different variations of the RPS game.
In the first variation (RPS4) we will consider four states $\{0, 1, 2, 3\}$ and the rules of the game will be $i+1$ beats i mod 4 for $i, j \in \{0, 1, 2, 3\}$. That is, 0 beats 3, 3 beats 2, 2 beats 1 and 1 beats 0. The rest of the interactions will be considered as a draw. This model is studied to see if changing the number of states changes the dynamics. It will also exemplify how non-interactions (draws) affect the system.

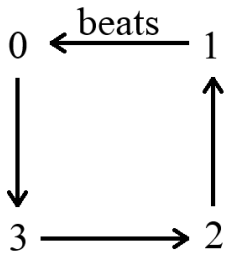

Figure 2: Diagram representing the 4 states variation of RPS.

The other variation is called Rock-Paper-Scissors-Lizard-Spock (RPSLS) and consists of 5 states {Rock, Paper, Scissors, Lizard, Spock}, adding lizard and Spock[1] to the already preexisting states in RPS. RPSLS was invented by Sam Kass and later popularized by the television show "The Big Bang Theory". Unlike the 4 states version, all the states interact with each other. The interactions can be represented by the complete directed graph 3.

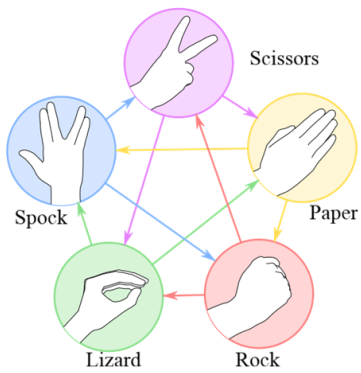

Figure 3: Diagram representing the rules of RPSLS. A good explanation of the rules can be found here.

This extension of the game allows for more complex interactions and therefore different patterns.

[1] Fictional character in Star Trek

3 Implementation as CA's

To model how these games evolve, we will use cellular automatas. In all three variations we will generate a $N \times N$ grid size where each cell contains a state. The grid will be initialized randomly.

The whole grid is updated simultaneously for each generation. To update the value in one cell, we define a threshold variable, which will remain fixed throughout the whole simulation.
Let's first consider the case where each state only has 1 predator, that is, models RPS, and RPS4. Let $X^t_{i,j} = s$ be the state of the cell (i, j) at a time t and p be the predator of s. We will count the number of predators of s in the neighborhood of (i, j) using a Moore's neighborhood (the 8 immediate neighbors). If the number of predators in the neighborhood exceeds the threshold value, we will update the cell to state p, that is $X^{t+1}_{i,j} = p$, if not, the cell will remain in the same state. Take for example figure 4, where yellow represents scissors, red paper and green rock. The cell in the middle is scissors, and if we set the threshold to 3, as we see 3 rocks in it's neighborhood the state of the cell in the middle will become a rock in the next generation.

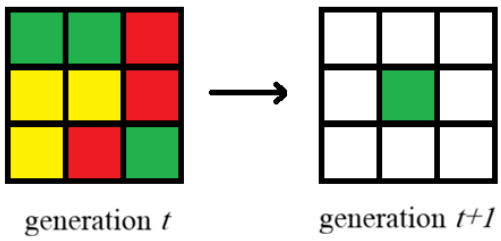

Figure 4: Example of update in the RPS model. Yellow is scissors, red is paper and green is rock.

In the RPSLS model the update isn't so trivial. This is due to having more than one predator per state. Take for example figure 5 and let blue be Spock. Recall that the predators of scissors (yellow) are rock (green) and Spock (blue). If we set the threshold to 2, both Spock and rock beat scissors, and both would be able to take over the cell, so we need a way to decide how to update it when conflict between predators arises. A way to remedy this is by adding priority proportional to the number of predators of each type. In example 5 more Spocks are in the neighborhood than rocks, so we could decide to update it to Spock. However, we would still have the problem on how to update it in case of draw. We decided to implement a random choice so no draws can happen. Therefore, if two predators are able to take over a cell, we will choose randomly which one takes over, regardless of how many predators are in the neighborhood.

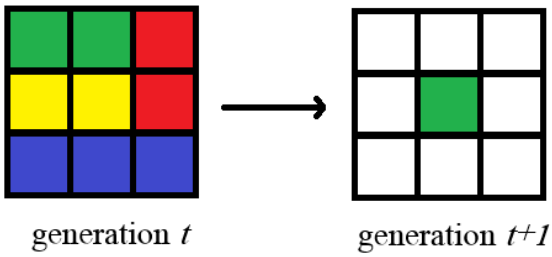

Figure 5: Example of update in the RPSLS model when conflict arises. Cell can become either green or blue; the choice is random.

The update of the boundary conditions will be done using periodic boundary conditions, resulting in a toroidal arrangement.

4 Results and Discussion

We implemented the model in python and plotted the simulation via an animation of the evolution of the system. The scripts are available at the project repository[2]. We will now present some results and observations from these animations. If not stated otherwise, the threshold value is set to 3 and the grid size to 200×200.

4.1 Waves and spirals

Although the simulation is initialized randomly we can see how it quickly self-organizes into spiral and wave patterns. This spirals represent three species competing in an area of the grid and account for the main dynamics of the system. The spirals propagate in a wave form and collide with other spirals, which lead to complex behaviors and patterns.

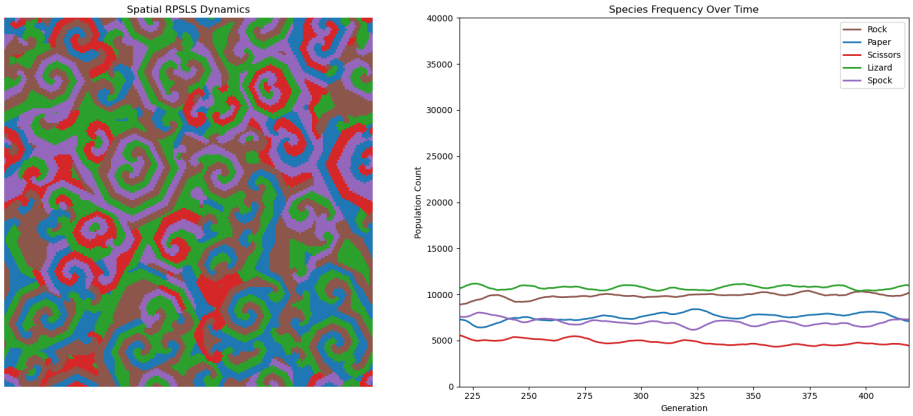

Figure 6: Spiral-like patterns in the RPSLS mdoel.

A key observation is that spirals are more frequently composed of three colors, although it is possible for 4 and 5 color spirals to be present. This makes it so spirals are harder to form in the RPS4 model, where states tends to propagate in larger blocks.

The propagation being of a wave form can be explained with the following diagram, where we suppose the pattern extends diagonally.

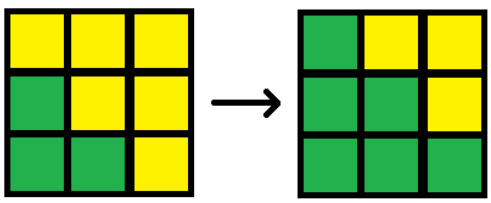

Figure 7: Wave propagation of a species. It is assumed that this pattern is extended diagonally. Each yellow cell in the boundary has 3 green neighbors and therefore transforms into a green cell.

The wave continues until it is stopped by an obstacle (usually another wave or spiral). This pattern can be repeating due to the cyclic nature of competition, as the green species could be chased down by it's predator.
Spirals are formed because of waves propagating in a circular motion, a species at the edge of the spiral is chased by one, while it chases another. These spirals promote biodiversity as they prevent one species from dominating the entire grid by constantly generating an equal amount of each species that constitutes

[2]https://github.com/edugoyu/Rock-Paper-Scissors-Dynamics-in-a-2-Dimensional-Cellular-Automata

the spiral. This behavior is local, meaning that multiple spirals are active simultaneously in the grid and only interact once they collide among each other. As time progresses we can actually see the spirals more defined and separated.

Figure 8: Time evolution of the grid in the RPSLS model. The spirals become clearer and more separated as time progresses. They maintain biodiversity by being constant sources of species.

4.2 Dependence of the threshold value

In all models, the patterns emerging are very dependent on the threshold value.
When setting the threshold to 2, the RPS model behaves in a very chaotic way and no pattern is recognizable. The image becomes very grainy and random. This is due to the high chance of a cell changing it's state, as a lower threshold can be interpreted as relaxing the conditions for a cell to change state. Since the initialization is random the updates behave in an almost random way and no structure can be seen. However, all frequencies stabilize around the same value.

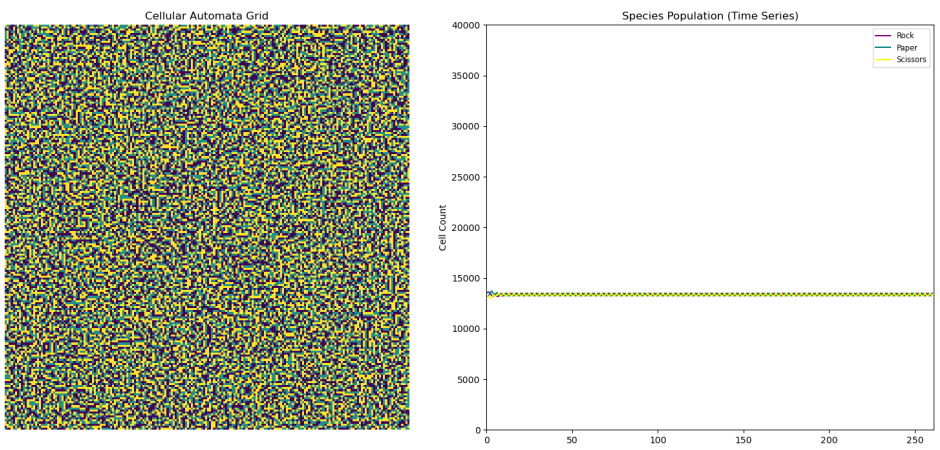

Figure 9: Snapshot of the RPS model with a threshold of 2.

When increasing the number of states, the chances of changing state become lower and patterns start emerging after a few generations. This can be seen in the RPS4 and RPSLS model, where wave and spiral patterns emerge and occupy the whole grid after few generations.

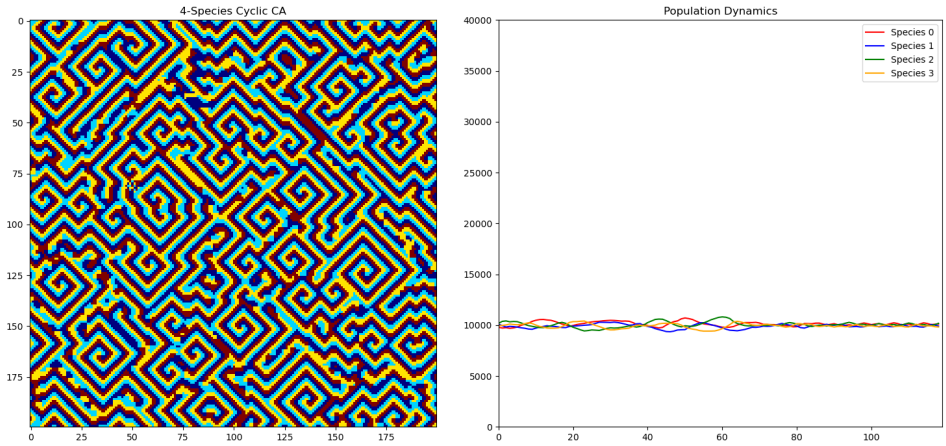

Figure 10: Snapshot of the RPS4 model with a threshold of 2.

However, one thing that distinguishes the RPS4 and RPSLS model when the threshold is 2 is the evolution of the frequencies of each state. In the RPS4 model all states stabilize around the same frequency, while in the RPSLS model the frequencies stabilize in an unequal way, that is, one or more states tend to dominate.

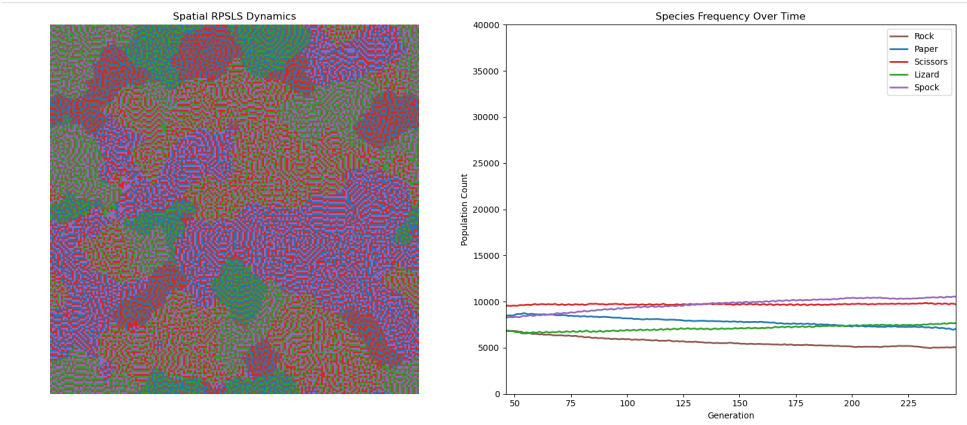

Figure 11: Snapshot of the RPSLS model with a threshold of 2.

When increasing the threshold to 3, all models have patterns and larger blocks of one species are created and destroyed, making so the frequencies behave in a sinusoidal way. The patterns mentioned before now emerge also in the RPS model and movement becomes slower and less chaotic in general.
Since blocks of a single state are now bigger, in the RPS4 model, due to non-interactions between states we can occasionally see "oasis" of a state in the middle of a block of another state, which last until perturbed by another state. This can happen for example in a block of state 2 and inside a group of cells in state 0.

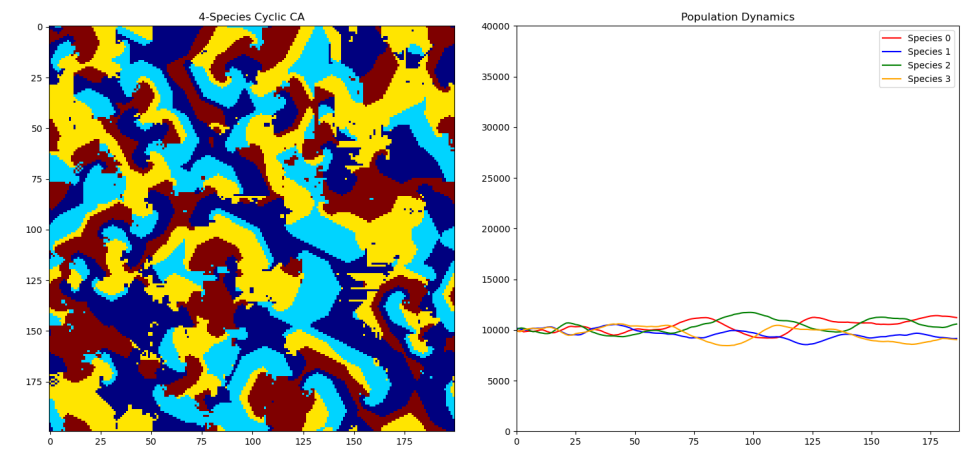

Figure 12: Snapshot of the RPS4 model with a threshold of 3. In the image one can see several oasis of a species within regions of another species.

Increasing the threshold past 4 freezes the system in a configuration where no updates happen, as it becomes increasingly less probable for a cell to change state. This becomes a snowball effect and eventually leads to the system freezing.
We can thus interpret the threshold value as being inversely proportional to the speed the system changes at, acting as a damper.

4.3 Dependence on the size of the population

We found that the size of the grid is also an important factor that determines the dynamics of the system. In a small grid size of 20×20 one of two things can happen; it can either become a periodic solution or one species can eventually dominate and extinguish the others.
In smaller systems, changes in frequency are more noticeable and are more likely to disrupt the balance between species, making it more probable for a species to extinguish. This creates an unbalance in the system, where one species gets rid of one of its predators and becomes more likely to dominate. However, we did find that the RPSLS model tends to last longer before one species dominates.
This exemplifies the importance of the balance of species in the system, as removing a predator to a state gives it an advantage over the others. This can also explain how in the RPSLS model, states tend to extinguish in pairs, and the model transforms into a RPS model, a balanced model.

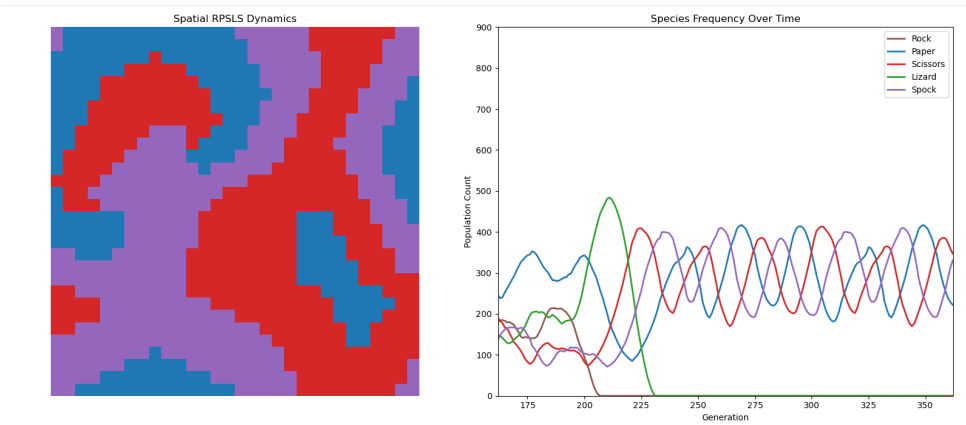

Figure 13: 20×20 grid of the RPSLS model. Lizard and Spock extinguish almost simultaneously and it becomes a RPS model.

Periodic solutions become more frequent in smaller grid sizes as with fewer components, the system returns to a state it has visited before with a bigger probability.
Increasing the size of the grid makes the fluctuations in frequency less noticeable, so the system doesn't die out. It also reduces the possibility of the simulation to become periodic. However, as mentioned in

section 4.2, frequencies oscillate around the statistical equilibrium, $r = \frac{N^2}{S}$ where S is the number of different species. In plot 14 we ran the simulation and plotted the phase space of the frequencies of two species (rock and paper). We see how the values oscillate around (r, r) but not in a purely periodic way.

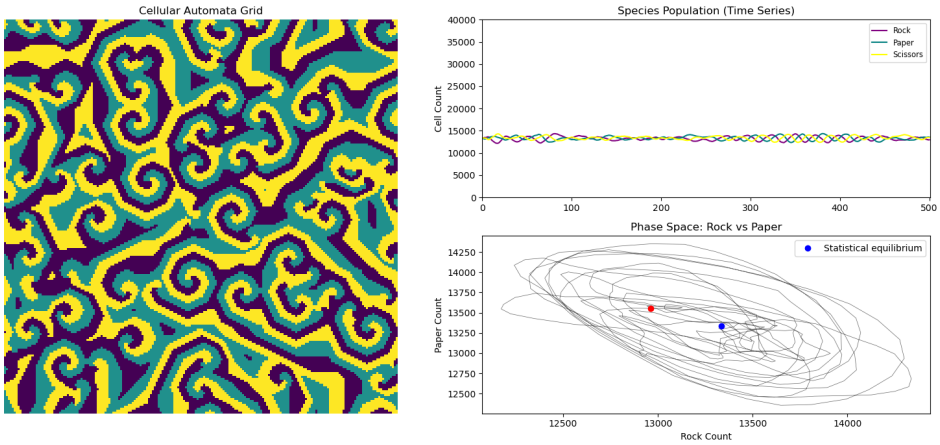

Figure 14: Almost periodic oscillations in the frequencies between rock and paper in the RPS model.

In general, biodiversity and coexistence in cyclic competition models are correlated with the population size. Specifically, high population size promotes biodiversity while in a small population it is more probable for the system to become periodic or fall into an absorbing state (dominance of a species) [3].

5 Conclusions

In this research, we have studied the dynamics of cyclical models of competition using the classical Rock-Paper-Scissors game and two other variants. We have seen that cellular automata provide a good framework to model these dynamics and found that complex patterns and behaviors can emerge from simple rules.

We've observed and discussed the influence different parameters have in the dynamics of the model. Particularly, how size and update rules determine the evolution of the frequencies of each species.
The value of the threshold can be interpreted as acting as a damper in the system. A small threshold of 2 makes the image grainier and doesn't allow blocks of one species to form while a bigger value of 4 eventually makes the simulation static. Because of this, we found that the most interesting value to study was 3.
In general, bigger population sizes tend to have richer dynamics than smaller ones, and are more prone to generate spiral and wave patterns. They also promote biodiversity by reducing the impact of stochastic fluctuations. Small populations are more likely to be affected by the oscillations of frequencies which can eventually lead to one species extinguishing and an unbalance in the system.

Studying these dynamics matter as they can model behaviors seen in nature. Searching patterns and explaining behaviors leads to further understanding on the biodiversity and stability of cyclical competition models. Therefore, these models provide a robust theoretical foundation and emphasize that the survival of multiple species is a scale-dependent process and the importance of balance between species. Further work can be done by exploring these relations in a deeper way. Research on how spirals form and why they become bigger as the system evolves could be studied. One could also explore the relation between the size of the grid and the formation of spirals, as we have seen that both are essential to explain the biodiversity of the system.

References

[1] Barry Sinervo and Curtis Lively. "The rock-paper-scissors game and the evolution of alternative male strategies". In: *Nature* 380 (Mar. 1996), pp. 240–243. DOI: 10.1038/380240a0.

[2] Jeffery Merkens et al. "The Dynamics of a Spatial Cyclic Competition System". In: *Technical report of the Mathematical and Theoretical Biology Institute* (Jan. 2009).

[3] Jens Christian Claussen and Arne Traulsen. "Cyclic Dominance and Biodiversity in Well-Mixed Populations". In: *Physical Review Letters* 100.5 (Feb. 2008). ISSN: 1079-7114. DOI: 10.1103/physrevlett.100.058104. URL: http://dx.doi.org/10.1103/PhysRevLett.100.058104.